Organic Acids

Carboxylic acids

Carboxylic Acids

A carboxylic acid

- Contains a *carboxyl group,* which is a carbonyl group (C=O) attached to a hydroxyl group (—OH).
- Has the carboxyl group on carbon 1.

carbonyl group

$$\begin{array}{c} O \\ \| \\ CH_3 — C — OH \end{array}$$

hydroxyl group or CH_3**COOH**

carboxyl group

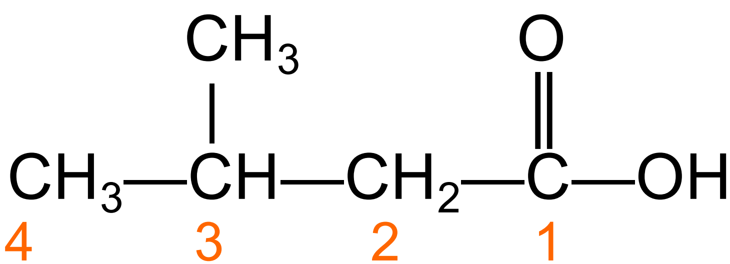
IUPAC Names

The IUPAC names of carboxylic acids

- Replace the -*e* in the alkane name with -*oic acid*.

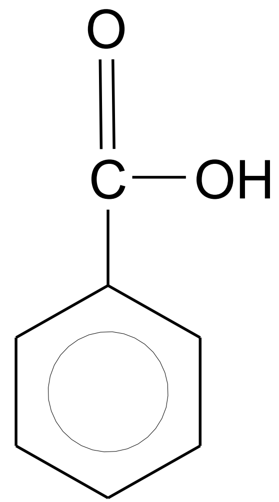
CH_4	methane	HCOOH	methanoic acid
CH_3—CH_3	ethane	CH_3—COOH	ethanoic acid

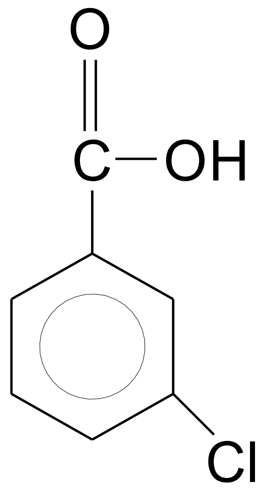
- Number substituents from the carboxyl carbon 1.

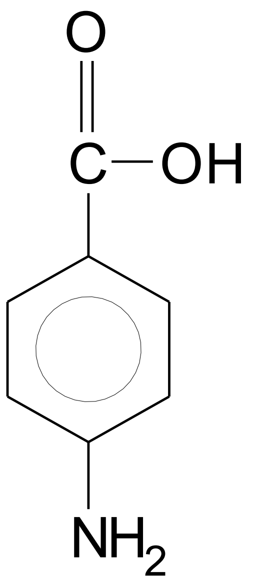


3-methylbutanoic acid

Aromatic Carboxylic Acids







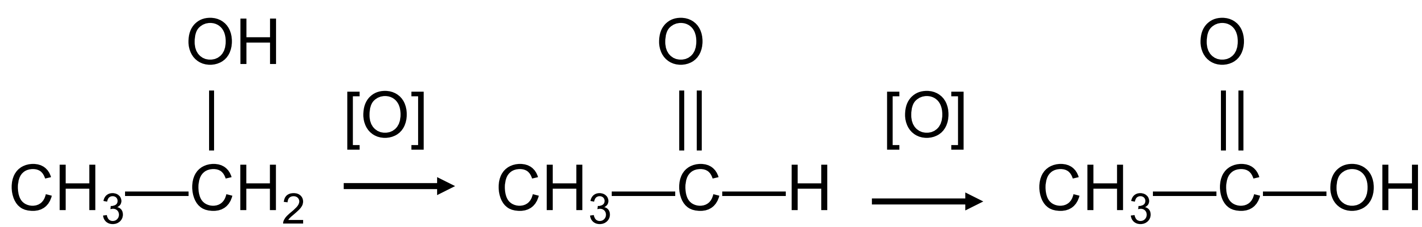
benzoic acid

3-chlorobenzoic acid
m-chlorobenzoic acid

4-aminobenzoic acid
p-aminobenzoic acid

Preparation of Carboxylic Acids

- Carboxylic acids can be prepared by oxidizing primary alcohols or aldehydes.
- The oxidation of ethanol produces ethanoic acid (acetic acid).



ethanol (ethyl alcohol)	ethanal (acetaldehyde)	ethanoic acid (acetic acid)

Properties of Carboxylic Acids

Acidity of Carboxylic Acids

Carboxylic acids

- Are weak acids.
- Ionize in water to produce carboxylate ions and hydronium ions.

$$CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-OH + H_2O \rightleftharpoons CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-O^- + H_3O^+$$

Physical properties of carboxylic acid

1-The first four acids formic to butyric are completely soluble in water

2-The polarity due to the carboxyl group is evident in the boiling point data formic acid boils at about 101c and the boiling point of acetic acid is 118c comparatively high boiling point for carboxylic acids due to intermolecular attractions result from hydrogen bonding.

3- Carboxylic acids are common in biochemistry when these molecules ionize, both the carboxylate anion and the (H^+) can have significant effects, for example if the liver releases too many carboxylic acids the blood become too acidic. This happens during uncontrolled diabetes and the resulting ketoacidosis can be deadly.

Medical importance of some organic acids

1- Formic acid (Methanoic acid) HCOOH

It **is a colorless liquid with a sharp irritating odor. Formic acid is found in the sting of bees and ants, causes a pain and swelling when it is injected into the tissues. it is one of the strongest organic acids.**

2- Acetic acid (Ethanoic acid) $CH3COOH$

It is one of the components of vinegar, where it is usually found as 4-5%solution. Acetic acid freeze at 17c, for that called glacial acetic acid. acetic acid can be made by the oxidation of ethanol.

3-Lactic acid

Lactic acid contains both a carboxyl group and a hydroxyl group, and is called a hydroxy acid. lactic acid is found in sour milk. it is found in the fermentation of milk sugar.

4-Oxalic acid

Is the simplest dicarboxylic acid, have the two carboxyl group. it is poisonous when taken internally, oxalate salts also prevent clotting by removing Ca ions from the blood.

5-Tartaric acid

Tartaric acid is found in several fruits, particularly grapes, potassium hydrogen tartarate, an acid salt called cream of tartar, is used in making baking powders. potassium sodium tartarate, known as Rochelle salts is used as a mild Cathartic.

6- Salicylic acid

Salicylic acid; is both a carboxylic acid and phenol it is of special interest because a family of useful drugs- the salicylate are derivatives of the acid, the salicylates include aspirin and function as analgesics and antipyretics.

7- para-Aminobenzoic acid (PABA)

It is also required by bacteria for the production of folic acid, needed to maintain the growth of healthy cell walls; sulfa drugs block the uptake of PABA by bacteria, causing them to be unable to manufacture folic acid, and thus preventing the bacteria from multiplying.

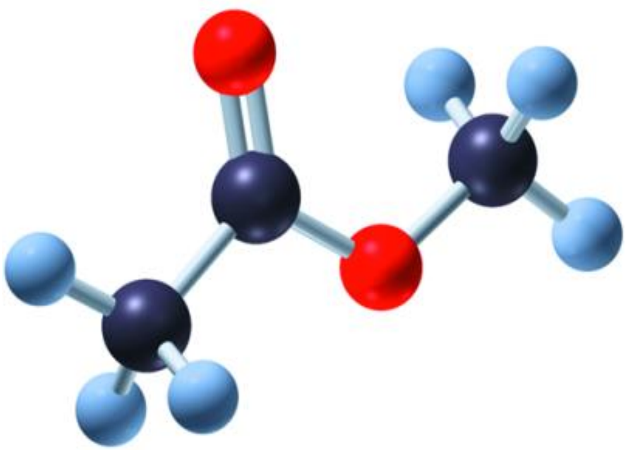
Esters

In an ester,

- The H in the carboxyl group is replaced with an alkyl group.

$$\mathrm{CH_3 - \overset{\displaystyle O}{\overset{\|}{C}} - O - CH_3}$$

ester group



Esters in Plants

Esters give flowers and fruits their pleasant fragrances and flavors.

Esterification

Esterification is

- The reaction of a carboxylic acid and alcohol in the presence of an acid catalyst to produce an ester.

$$CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-OH \;+\; H-O-CH_2-CH_3 \;\xrightleftharpoons{H^+}$$

$$CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-O-CH_2-CH_3 \;+\; H_2O$$

ethyl acetate (an ester)

Physical properties of Esters

Esters are more polar than ethers but less polar than alcohols. Consequently esters are more volatile than carboxylic acids of similar molecular weight.

Acid Hydrolysis of Esters

In acid hydrolysis

- An ester reacts with water to produce a carboxylic acid and an alcohol.
- An acid catalyst is required.

$$\mathrm{H{-}\overset{\overset{\displaystyle O}{\|}}{C}{-}O{-}CH_2{-}CH_3} + \mathrm{H_2O} \xrightarrow{\mathrm{H^+}}$$

$$\mathrm{H{-}\overset{\overset{\displaystyle O}{\|}}{C}{-}OH} + \mathrm{H{-}O{-}CH_2{-}CH_3}$$

Base Hydrolysis (Saponification)

Base hydrolysis (also called saponification)

- Is the reaction of an ester with a strong base.
- Produces the salt of the carboxylic acid and an alcohol.

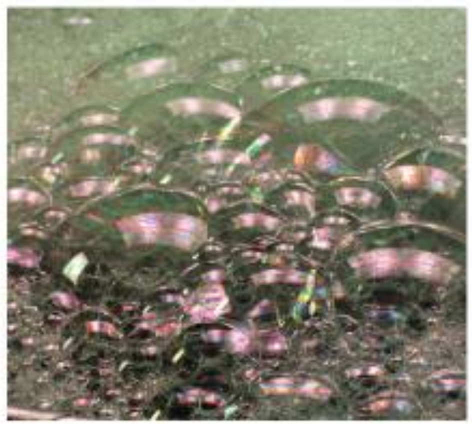
$$CH_3\text{—}\overset{\overset{\displaystyle O}{\|}}{C}\text{—}O\text{—}CH_2\text{—}CH_3 \;+\; NaOH \longrightarrow$$

$$CH_3\text{—}\overset{\overset{\displaystyle O}{\|}}{C}\text{—}O^-\,Na^+ \;+\; HO\text{—}CH_2\text{—}CH_3$$

salt of carboxylic acid　　　　　alcohol

“Soaps”

The base hydrolysis of long chain fatty acids produces acid salts called “soaps”.



Fatty acid

$$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2{-}\overset{\overset{\displaystyle O}{\|}}{C}{-}OH + NaOH \longrightarrow$$

Carboxylic acid salt, “soap”

$$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2{-}\overset{\overset{\displaystyle O}{\|}}{C}{-}O^- Na^+$$

Nonpolar tail — Polar head

Cleaning Action of Soap

A soap

- Contains a nonpolar end that dissolves in nonpolar

- and oils, and a polar end that dissolves in water.
- Forms groups of soap molecules that dissolve in water and are washed away.